Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by total mortality class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (104 sites, 381 site-years) · deadtrees v4.1

ΔRMSE = RMSE(with M) − RMSE(without M)

	XGBoost (Optuna) · 12m	XGBoost (Optuna) · 24m	LightGBM (Optuna) · 12m	LightGBM (Optuna) · 24m
GPPsat (µmol m⁻² s⁻¹)	66% improved	56% improved	63% improved	58% improved
NEPmax (µmol m⁻² s⁻¹)	76% improved	70% improved	73% improved	74% improved
ETmax (mm d⁻¹)	56% improved	61% improved	54% improved	54% improved
WUE (g C mm⁻¹)	53% improved	55% improved	54% improved	56% improved

Sites (sorted by ΔRMSE within each panel)

Total mortality
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 12.4 %, 40 sites)
- Mid (12.4–25.0 %, 43 sites)
- High (> 25.0 %, 21 sites)